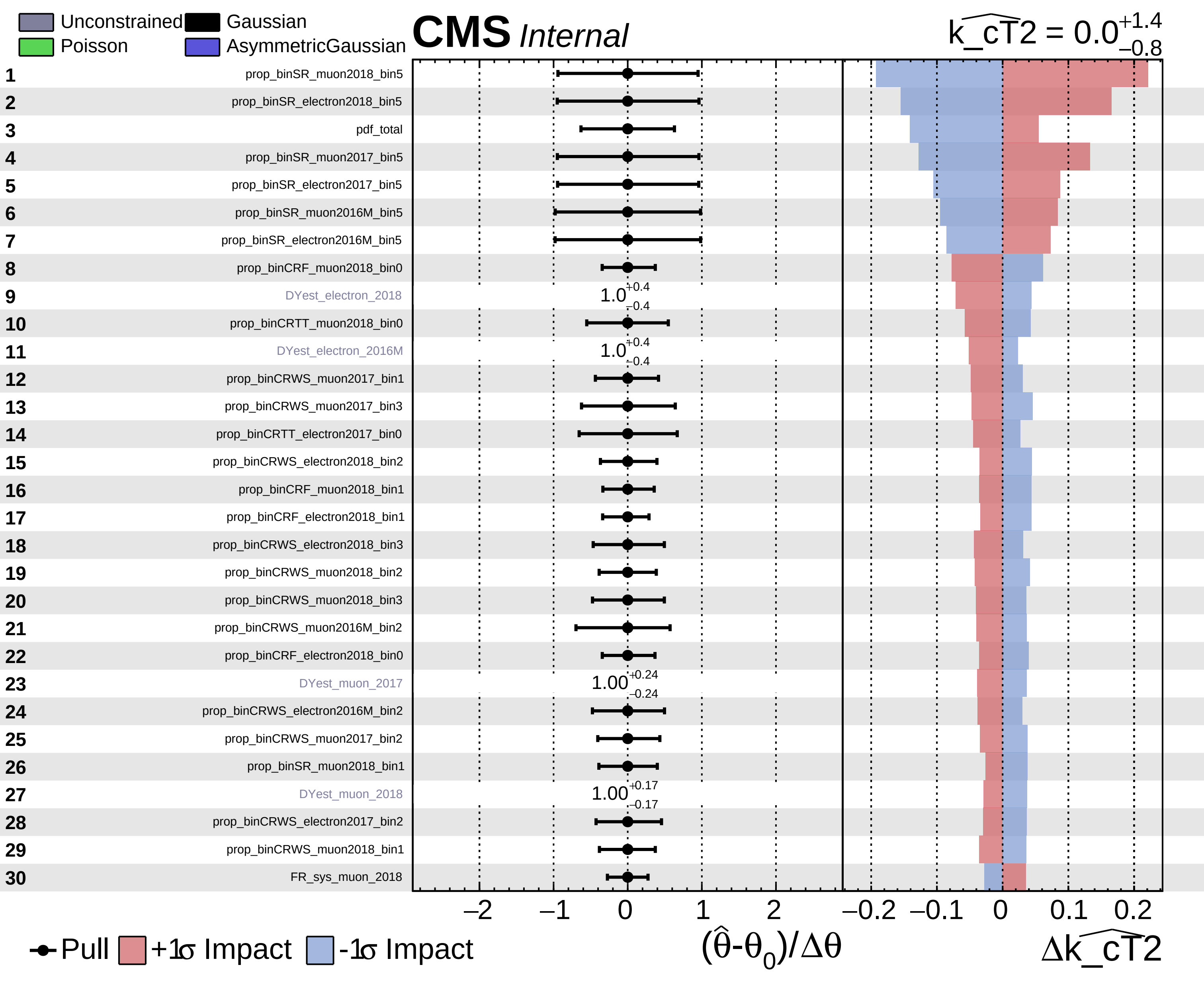
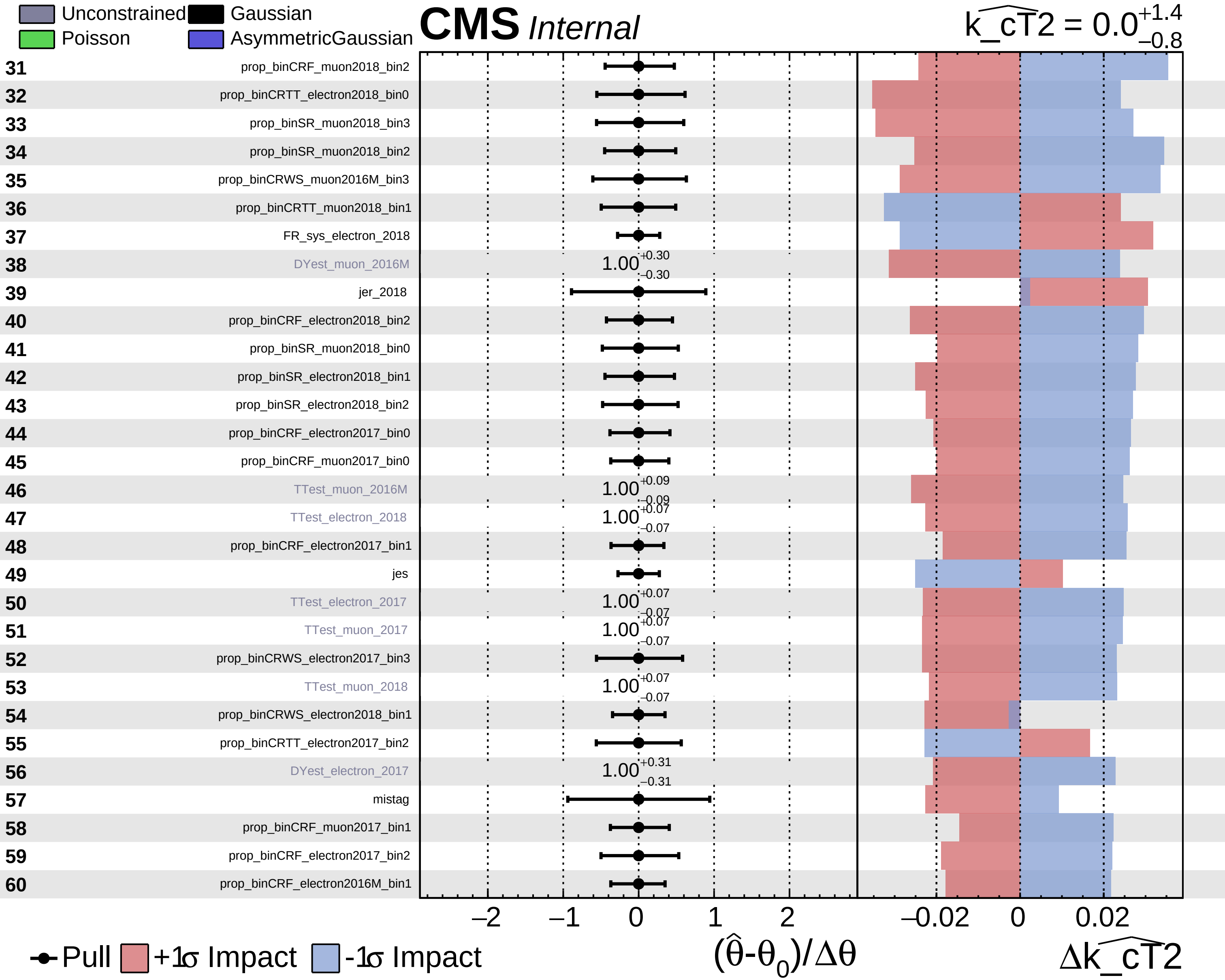
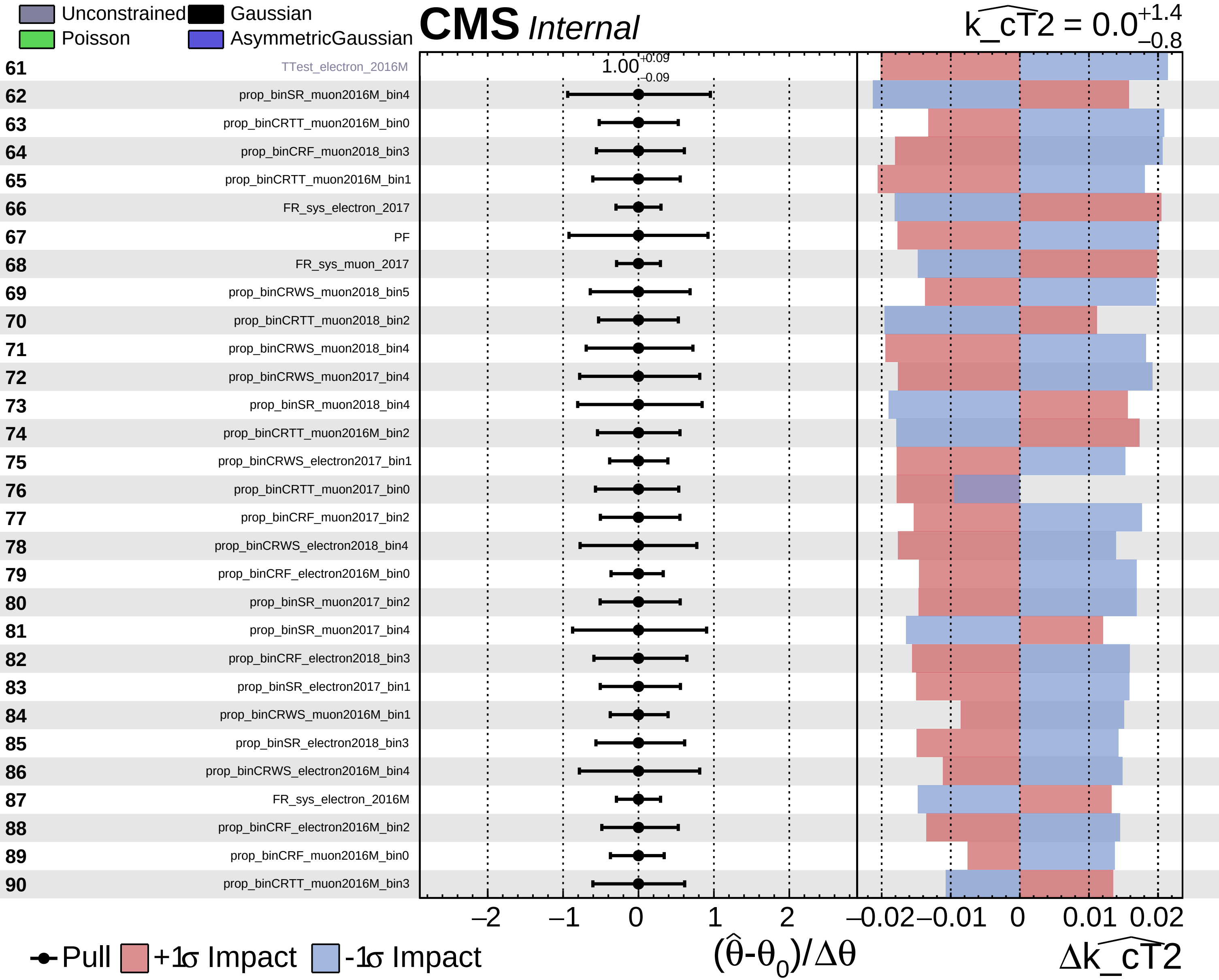
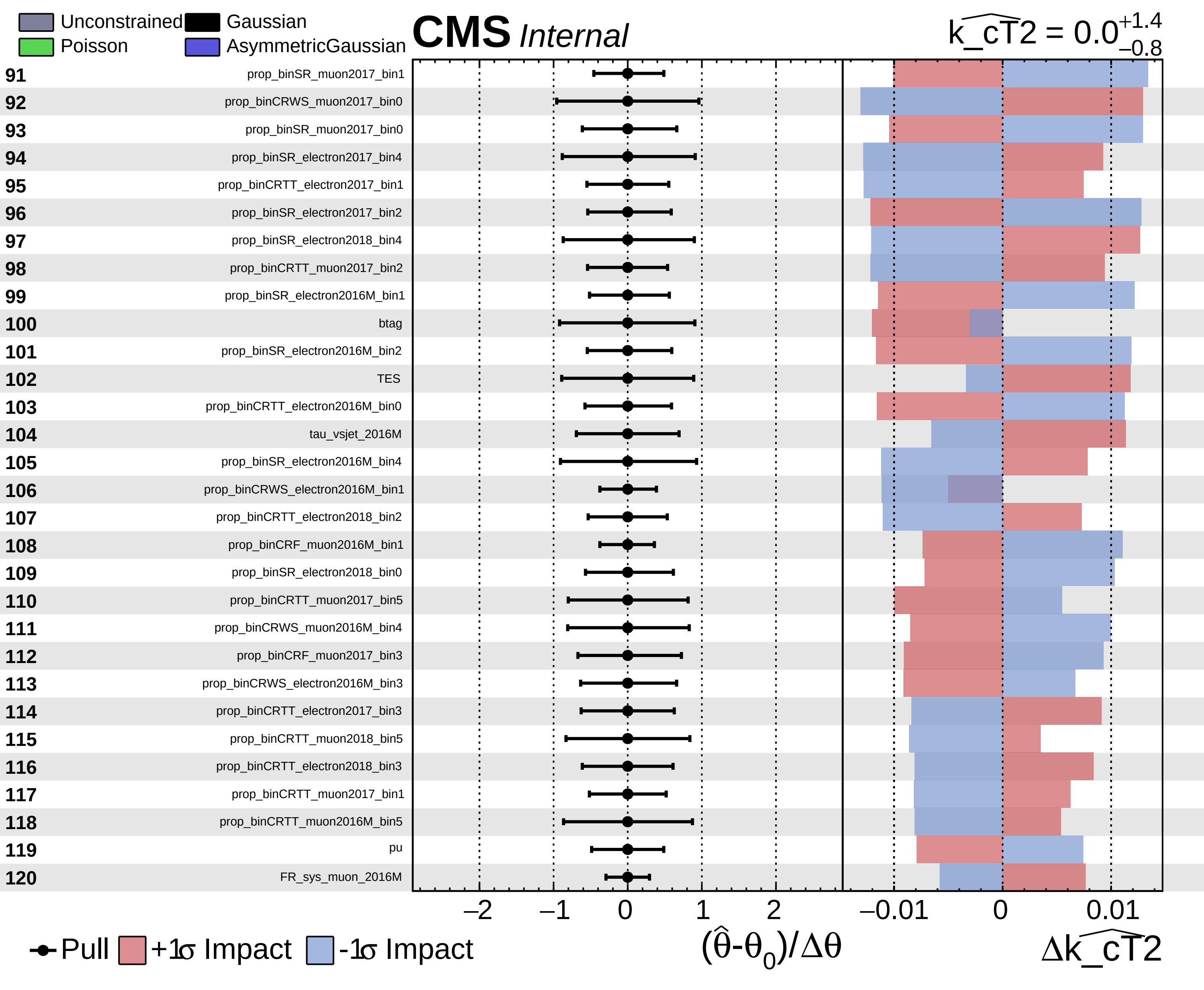


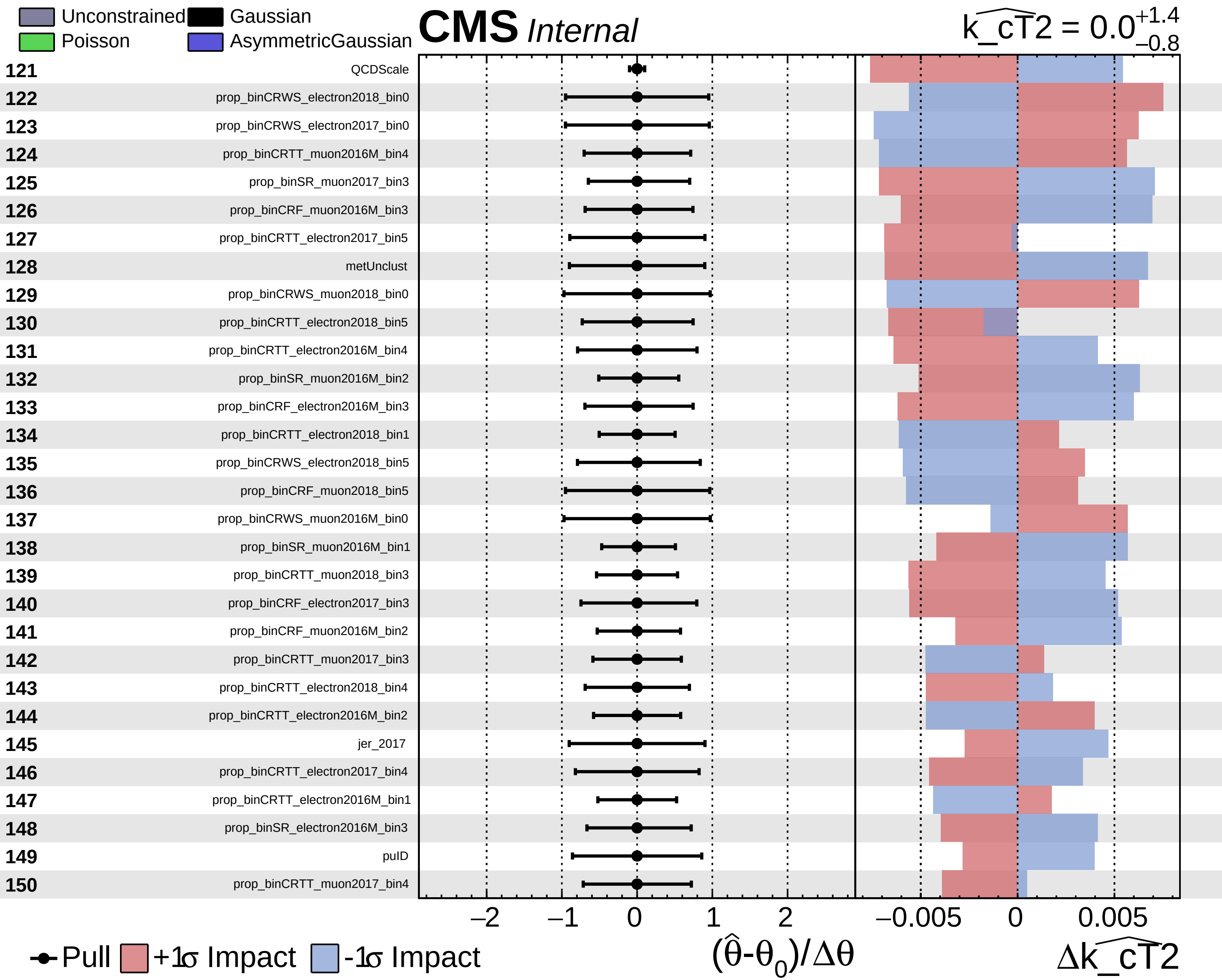
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cT2 = 0.0 +1.4 −0.8
1 prop_binSR_muon2018_bin5
2 prop_binSR_electron2018_bin5
3 pdf_total
4 prop_binSR_muon2017_bin5
5 prop_binSR_electron2017_bin5
6 prop_binSR_muon2016M_bin5
7 prop_binSR_electron2016M_bin5
8 prop_binCRF_muon2018_bin0
9 DYest_electron_2018
1.0 +0.4 −0.4
10 prop_binCRTT_muon2018_bin0
11 DYest_electron_2016M
1.0 +0.4 −0.4
12 prop_binCRWS_muon2017_bin1
13 prop_binCRWS_muon2017_bin3
14 prop_binCRTT_electron2017_bin0
15 prop_binCRWS_electron2018_bin2
16 prop_binCRF_muon2018_bin1
17 prop_binCRF_electron2018_bin1
18 prop_binCRWS_electron2018_bin3
19 prop_binCRWS_muon2018_bin2
20 prop_binCRWS_muon2018_bin3
21 prop_binCRWS_muon2016M_bin2
22 prop_binCRF_electron2018_bin0
23 DYest_muon_2017
1.00 +0.24 −0.24
24 prop_binCRWS_electron2016M_bin2
25 prop_binCRWS_muon2017_bin2
26 prop_binSR_muon2018_bin1
27 DYest_muon_2018
1.00 +0.17 −0.17
28 prop_binCRWS_electron2017_bin2
29 prop_binCRWS_muon2018_bin1
30 FR_sys_muon_2018
−2 −1 0 1 2
−0.2 −0.1 0 0.1 0.2
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δk_cT2



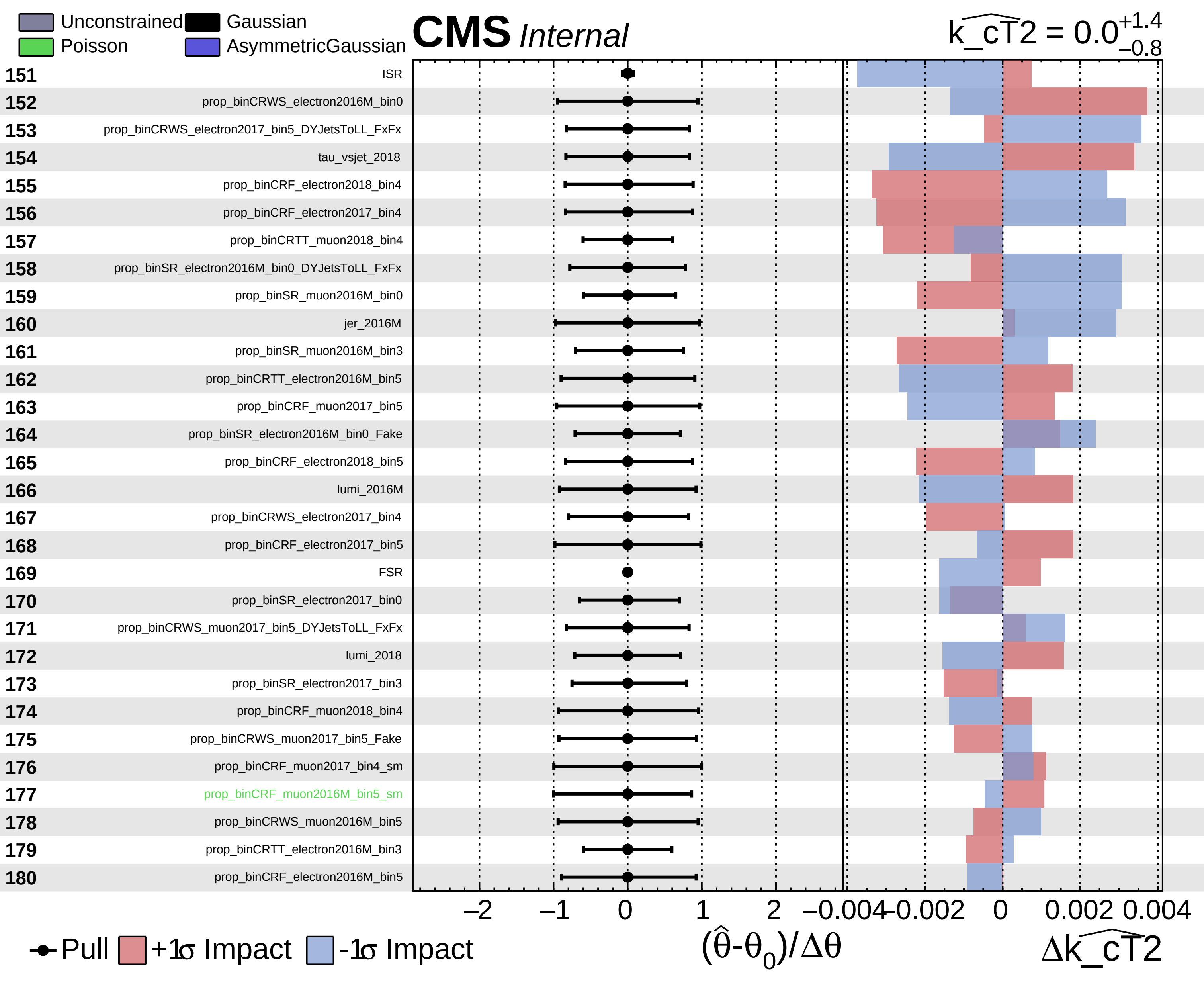
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cT2 = 0.0+1.4 −0.8
31 prop_binCRF_muon2018_bin2
32 prop_binCRTT_electron2018_bin0
33 prop_binSR_muon2018_bin3
34 prop_binSR_muon2018_bin2
35 prop_binCRWS_muon2016M_bin3
36 prop_binCRTT_muon2018_bin1
37 FR_sys_electron_2018
38 DYest_muon_2016M 1.00 +0.30 −0.30
39 jer_2018
40 prop_binCRF_electron2018_bin2
41 prop_binSR_muon2018_bin0
42 prop_binSR_electron2018_bin1
43 prop_binSR_electron2018_bin2
44 prop_binCRF_electron2017_bin0
45 prop_binCRF_muon2017_bin0
46 TTest_muon_2016M 1.00 +0.09 −0.09
47 TTest_electron_2018 1.00 +0.07 −0.07
48 prop_binCRF_electron2017_bin1
49 jes
50 TTest_electron_2017 1.00 +0.07 −0.07
51 TTest_muon_2017 1.00 +0.07 −0.07
52 prop_binCRWS_electron2017_bin3
53 TTest_muon_2018 1.00 +0.07 −0.07
54 prop_binCRWS_electron2018_bin1
55 prop_binCRTT_electron2017_bin2
56 DYest_electron_2017 1.00 +0.31 −0.31
57 mistag
58 prop_binCRF_muon2017_bin1
59 prop_binCRF_electron2017_bin2
60 prop_binCRF_electron2016M_bin1
−2 −1 0 1 2
−0.02 0 0.02
Pull +1σ Impact -1σ Impact
(θ̂-θ0)/Δθ
Δk_cT2

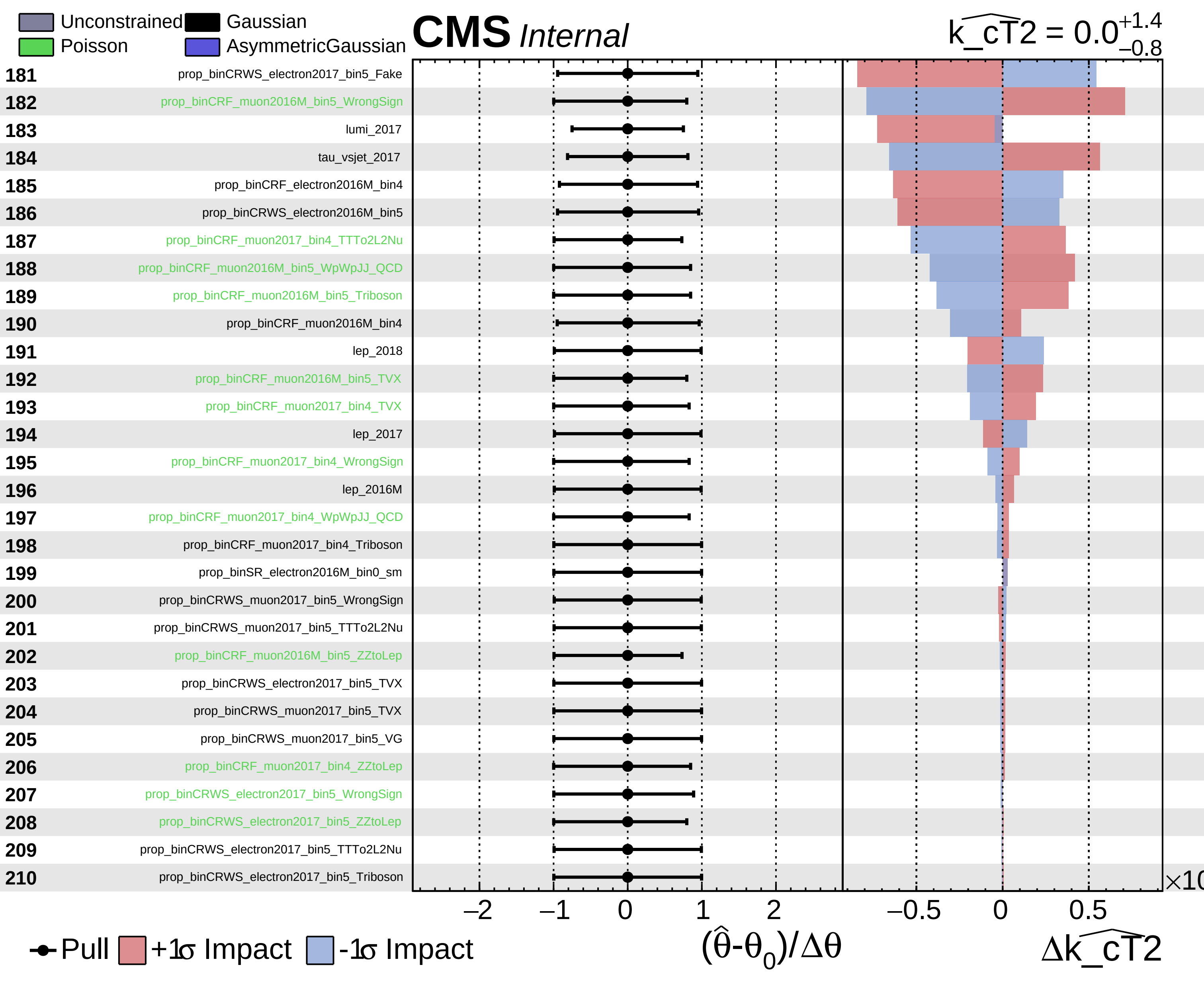

Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cT2 = 0.0 +1.4 −0.8
61 TTest_electron_2016M
1.00 +0.09 −0.09
62 prop_binSR_muon2016M_bin4
63 prop_binCRTT_muon2016M_bin0
64 prop_binCRF_muon2018_bin3
65 prop_binCRTT_muon2016M_bin1
66 FR_sys_electron_2017
67 PF
68 FR_sys_muon_2017
69 prop_binCRWS_muon2018_bin5
70 prop_binCRTT_muon2018_bin2
71 prop_binCRWS_muon2018_bin4
72 prop_binCRWS_muon2017_bin4
73 prop_binSR_muon2018_bin4
74 prop_binCRTT_muon2016M_bin2
75 prop_binCRWS_electron2017_bin1
76 prop_binCRTT_muon2017_bin0
77 prop_binCRF_muon2017_bin2
78 prop_binCRWS_electron2018_bin4
79 prop_binCRF_electron2016M_bin0
80 prop_binSR_muon2017_bin2
81 prop_binSR_muon2017_bin4
82 prop_binCRF_electron2018_bin3
83 prop_binSR_electron2017_bin1
84 prop_binCRWS_muon2016M_bin1
85 prop_binSR_electron2018_bin3
86 prop_binCRWS_electron2016M_bin4
87 FR_sys_electron_2016M
88 prop_binCRF_electron2016M_bin2
89 prop_binCRF_muon2016M_bin0
90 prop_binCRTT_muon2016M_bin3
−2 −1 0 1 2
−0.02 −0.01 0 0.01 0.02
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δk_cT2

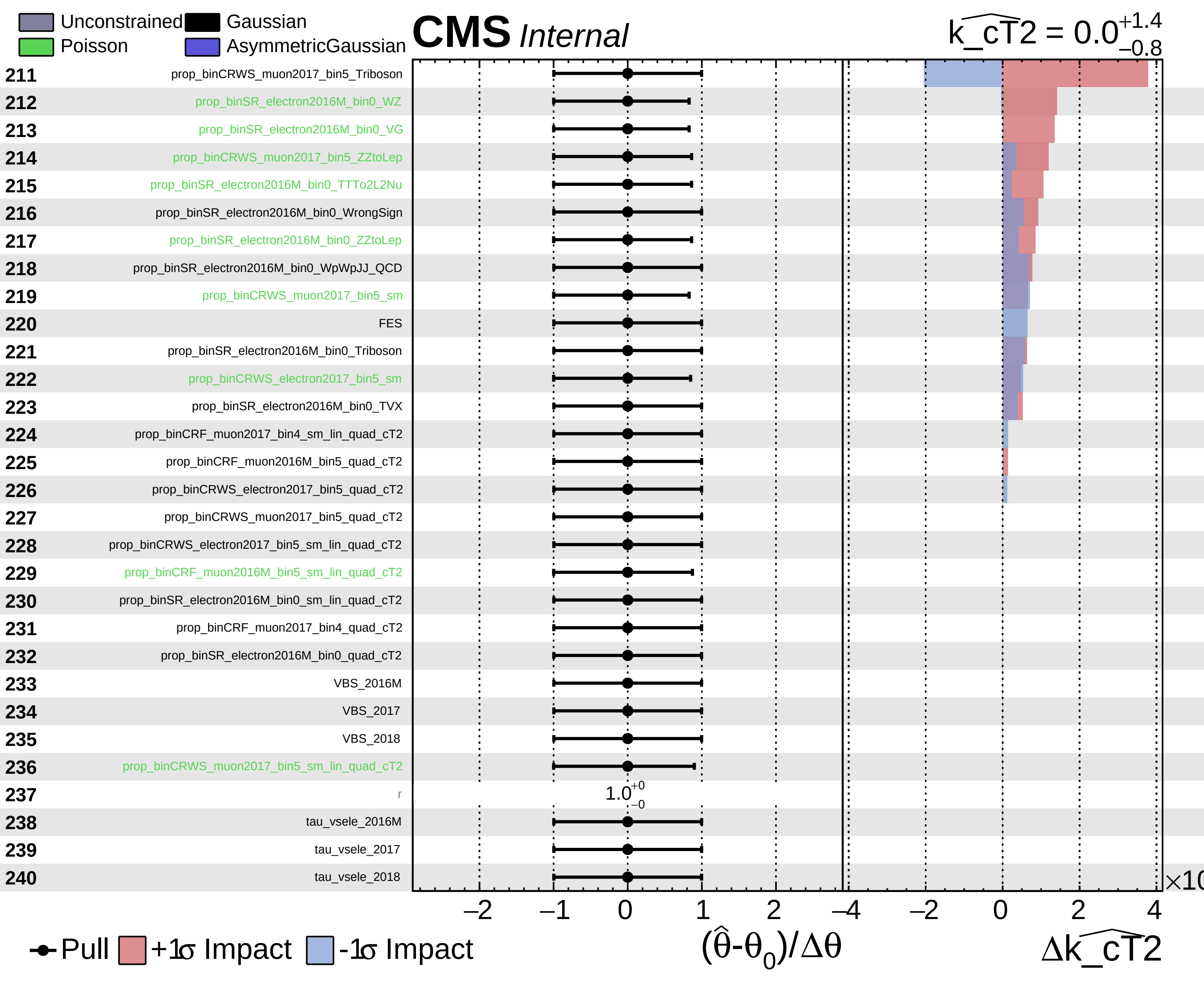

Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\widehat{k\_cT2} = 0.0^{+1.4}_{-0.8}$
91 prop_binSR_muon2017_bin1
92 prop_binCRWS_muon2017_bin0
93 prop_binSR_muon2017_bin0
94 prop_binSR_electron2017_bin4
95 prop_binCRTT_electron2017_bin1
96 prop_binSR_electron2017_bin2
97 prop_binSR_electron2018_bin4
98 prop_binCRTT_muon2017_bin2
99 prop_binSR_electron2016M_bin1
100 btag
101 prop_binSR_electron2016M_bin2
102 TES
103 prop_binCRTT_electron2016M_bin0
104 tau_vsjet_2016M
105 prop_binSR_electron2016M_bin4
106 prop_binCRWS_electron2016M_bin1
107 prop_binCRTT_electron2018_bin2
108 prop_binCRF_muon2016M_bin1
109 prop_binSR_electron2018_bin0
110 prop_binCRTT_muon2017_bin5
111 prop_binCRWS_muon2016M_bin4
112 prop_binCRF_muon2017_bin3
113 prop_binCRWS_electron2016M_bin3
114 prop_binCRTT_electron2017_bin3
115 prop_binCRTT_muon2018_bin5
116 prop_binCRTT_electron2018_bin3
117 prop_binCRTT_muon2017_bin1
118 prop_binCRTT_muon2016M_bin5
119 pu
120 FR_sys_muon_2016M
−2 −1 0 1 2
−0.01 0 0.01
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cT2}$

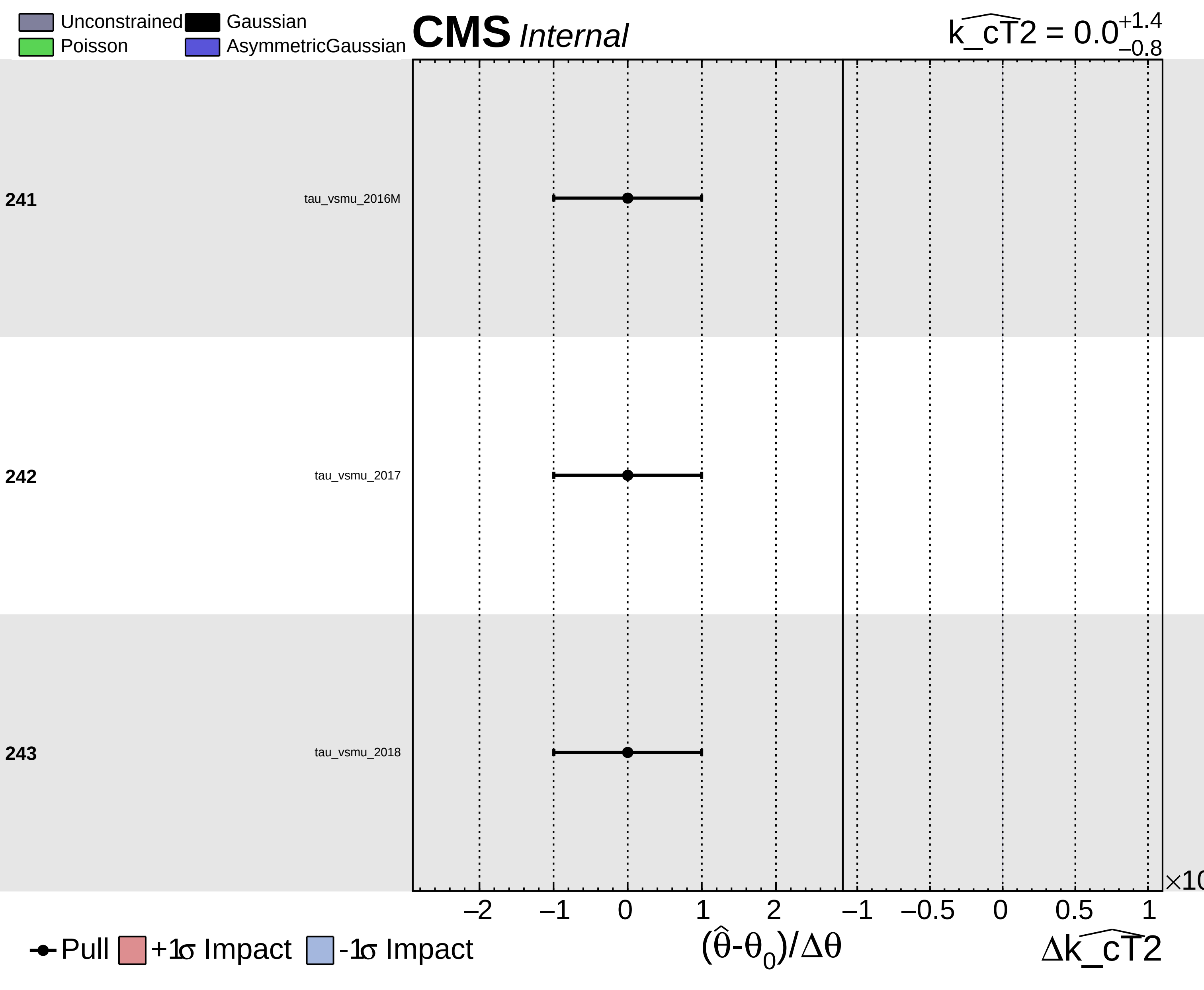


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\widehat{k\_cT2} = 0.0^{+1.4}_{-0.8}$
121 QCDScale
122 prop_binCRWS_electron2018_bin0
123 prop_binCRWS_electron2017_bin0
124 prop_binCRTT_muon2016M_bin4
125 prop_binSR_muon2017_bin3
126 prop_binCRF_muon2016M_bin3
127 prop_binCRTT_electron2017_bin5
128 metUnclust
129 prop_binCRWS_muon2018_bin0
130 prop_binCRTT_electron2018_bin5
131 prop_binCRTT_electron2016M_bin4
132 prop_binSR_muon2016M_bin2
133 prop_binCRF_electron2016M_bin3
134 prop_binCRTT_electron2018_bin1
135 prop_binCRWS_electron2018_bin5
136 prop_binCRF_muon2018_bin5
137 prop_binCRWS_muon2016M_bin0
138 prop_binSR_muon2016M_bin1
139 prop_binCRTT_muon2018_bin3
140 prop_binCRF_electron2017_bin3
141 prop_binCRF_muon2016M_bin2
142 prop_binCRTT_muon2017_bin3
143 prop_binCRTT_electron2018_bin4
144 prop_binCRTT_electron2016M_bin2
145 jer_2017
146 prop_binCRTT_electron2017_bin4
147 prop_binCRTT_electron2016M_bin1
148 prop_binSR_electron2016M_bin3
149 puID
150 prop_binCRTT_muon2017_bin4
−2
−1
0
1
2
−0.005
0
0.005
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cT2}$


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\widehat{k\_cT2} = 0.0^{+1.4}_{-0.8}$
151 ISR
152 prop_binCRWS_electron2016M_bin0
153 prop_binCRWS_electron2017_bin5_DYJetsToLL_FxFx
154 tau_vsjet_2018
155 prop_binCRF_electron2018_bin4
156 prop_binCRF_electron2017_bin4
157 prop_binCRTT_muon2018_bin4
158 prop_binSR_electron2016M_bin0_DYJetsToLL_FxFx
159 prop_binSR_muon2016M_bin0
160 jer_2016M
161 prop_binSR_muon2016M_bin3
162 prop_binCRTT_electron2016M_bin5
163 prop_binCRF_muon2017_bin5
164 prop_binSR_electron2016M_bin0_Fake
165 prop_binCRF_electron2018_bin5
166 lumi_2016M
167 prop_binCRWS_electron2017_bin4
168 prop_binCRF_electron2017_bin5
169 FSR
170 prop_binSR_electron2017_bin0
171 prop_binCRWS_muon2017_bin5_DYJetsToLL_FxFx
172 lumi_2018
173 prop_binSR_electron2017_bin3
174 prop_binCRF_muon2018_bin4
175 prop_binCRWS_muon2017_bin5_Fake
176 prop_binCRF_muon2017_bin4_sm
177 prop_binCRF_muon2016M_bin5_sm
178 prop_binCRWS_muon2016M_bin5
179 prop_binCRTT_electron2016M_bin3
180 prop_binCRF_electron2016M_bin5
−2 −1 0 1 2
−0.004 −0.002 0 0.002 0.004
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cT2}$


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cT2 = 0.0 +1.4 −0.8
181 prop_binCRWS_electron2017_bin5_Fake
182 prop_binCRF_muon2016M_bin5_WrongSign
183 lumi_2017
184 tau_vsjet_2017
185 prop_binCRF_electron2016M_bin4
186 prop_binCRWS_electron2016M_bin5
187 prop_binCRF_muon2017_bin4_TTTo2L2Nu
188 prop_binCRF_muon2016M_bin5_WpWpJJ_QCD
189 prop_binCRF_muon2016M_bin5_Triboson
190 prop_binCRF_muon2016M_bin4
191 lep_2018
192 prop_binCRF_muon2016M_bin5_TVX
193 prop_binCRF_muon2017_bin4_TVX
194 lep_2017
195 prop_binCRF_muon2017_bin4_WrongSign
196 lep_2016M
197 prop_binCRF_muon2017_bin4_WpWpJJ_QCD
198 prop_binCRF_muon2017_bin4_Triboson
199 prop_binSR_electron2016M_bin0_sm
200 prop_binCRWS_muon2017_bin5_WrongSign
201 prop_binCRWS_muon2017_bin5_TTTo2L2Nu
202 prop_binCRF_muon2016M_bin5_ZZtoLep
203 prop_binCRWS_electron2017_bin5_TVX
204 prop_binCRWS_muon2017_bin5_TVX
205 prop_binCRWS_muon2017_bin5_VG
206 prop_binCRF_muon2017_bin4_ZZtoLep
207 prop_binCRWS_electron2017_bin5_WrongSign
208 prop_binCRWS_electron2017_bin5_ZZtoLep
209 prop_binCRWS_electron2017_bin5_TTTo2L2Nu
210 prop_binCRWS_electron2017_bin5_Triboson
−2 −1 0 1 2
−0.5 0 0.5
×10
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δk_cT2


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\widehat{k\_cT2} = 0.0^{+1.4}_{-0.8}$
211 prop_binCRWS_muon2017_bin5_Triboson
212 prop_binSR_electron2016M_bin0_WZ
213 prop_binSR_electron2016M_bin0_VG
214 prop_binCRWS_muon2017_bin5_ZZtoLep
215 prop_binSR_electron2016M_bin0_TTTo2L2Nu
216 prop_binSR_electron2016M_bin0_WrongSign
217 prop_binSR_electron2016M_bin0_ZZtoLep
218 prop_binSR_electron2016M_bin0_WpWpJJ_QCD
219 prop_binCRWS_muon2017_bin5_sm
220 FES
221 prop_binSR_electron2016M_bin0_Triboson
222 prop_binCRWS_electron2017_bin5_sm
223 prop_binSR_electron2016M_bin0_TVX
224 prop_binCRF_muon2017_bin4_sm_lin_quad_cT2
225 prop_binCRF_muon2016M_bin5_quad_cT2
226 prop_binCRWS_electron2017_bin5_quad_cT2
227 prop_binCRWS_muon2017_bin5_quad_cT2
228 prop_binCRWS_electron2017_bin5_sm_lin_quad_cT2
229 prop_binCRF_muon2016M_bin5_sm_lin_quad_cT2
230 prop_binSR_electron2016M_bin0_sm_lin_quad_cT2
231 prop_binCRF_muon2017_bin4_quad_cT2
232 prop_binSR_electron2016M_bin0_quad_cT2
233 VBS_2016M
234 VBS_2017
235 VBS_2018
236 prop_binCRWS_muon2017_bin5_sm_lin_quad_cT2
237 r
1.0^{+0}_{-0}
238 tau_vsele_2016M
239 tau_vsele_2017
240 tau_vsele_2018
−2 −1 0 1 2
−4 −2 0 2 4
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cT2}$


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cT2 = 0.0 +1.4 −0.8
241
tau_vsmu_2016M
242
tau_vsmu_2017
243
tau_vsmu_2018
×10
−2 −1 0 1 2
−1 −0.5 0 0.5 1
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δk_cT2